Product Context Document

Product: Internet Banking Concept Microservices
Repository: Persistent-Systems-Test/ts-java-spring-boot-internet-banking
Baseline: main @ fa1c445
Assessment date: 2026-11-03
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Product Context - Internet Banking Concept Microservices

This is the single source of truth for the PDD and TDD. Every statement is classified:

Class	Meaning
VERIFIED	Verified implementation - observed in source code/config in this repository
DOCUMENTED	Documented but unverified - stated in README/docs, not found in code
INFERRED	Inferred - assessor reasoning from code structure; not explicitly documented
MISSING	Missing information - not present in repository
STAKEHOLDER	Requires stakeholder validation - business information not derivable from code

1. Product overview and purpose

Aspect	Statement	Class	Evidence
Purpose	Digital (internet) banking channel platform exposing REST APIs for online-banking user onboarding, account lookup, intra-bank fund transfers and utility bill payments, on top of a simulated core banking ledger.	VERIFIED	E-01 `README.md` (1-20); E-26 `core-banking-service/src/main/java/com/javatodev/finance/controller/*.java`
Origin	Tutorial/reference implementation by javatodev.com (MIT), first commit 2021-05-09, upgraded to Java 21 / Spring Boot 3.2.4 in 2024.	VERIFIED	E-01 `README.md` (1-20); E-04 `LICENSE` (1-3); E-43 `git log`
Business Problem	Banks need a secure self-service digital channel decoupled from the core banking system so channel features evolve independently of the core.	INFERRED	E-01 `README.md` (1-20); E-26 `core-banking-service/src/main/java/com/javatodev/finance/controller/*.java`
Objectives	Stated objective is educational: demonstrate Spring Boot microservices patterns that 'evolve as a full fledged REST API collection until deployments'. Commercial objectives/outcomes are not documented.	DOCUMENTED / STAKEHOLDER	E-01 `README.md` (1-20)

2. Target users and stakeholders

User / stakeholder	Need / interaction	Class	Evidence
Retail bank customer (existing core customer)	Registers, logs in, transfers funds, pays bills	VERIFIED (API only, no UI)	E-28 `UserController.java` (28-55); E-33 `FundTransferService.java` (31-51); E-34 `UtilityPaymentService.java` (31-53)
Bank administrator / operations	Approves/blacklists registrations	VERIFIED (PATCH status; no role model)	E-28 `UserController.java` (28-55); E-29 `UserService.java` (33-111)
Developer / learner	Learns microservices patterns via tutorial, Postman, compose	DOCUMENTED	E-01 `README.md` (1-20); E-42 `JAVA_TO_DEV_MICROSERVICES.postman_collection.json`

User / stakeholder	Need / interaction	Class	Evidence
Compliance, risk, customer-service staff	No supporting capability	MISSING	No repository evidence available (absence of artefact)
Commercial buyer / target segment	Not documented	STAKEHOLDER	No repository evidence available (absence of artefact)

3. Functional capabilities

ID	Capability	Class	Evidence
CAP-01	Customer self-registration	VERIFIED	E-28 `UserController.java` (28-55); E-29 `UserService.java` (33-111)
CAP-02	Registration approval / status management	VERIFIED	E-28 `UserController.java` (28-55); E-29 `UserService.java` (33-111)
CAP-03	Authentication (OIDC/JWT via Keycloak)	VERIFIED	E-10 `realm-export.json` (686-873); E-15 `SecurityConfiguration.java` (24-35)
CAP-04	Bank account & biller lookup	VERIFIED	E-26 `core-banking-service/src/main/java/com/javatodev/finance/controller/*.java`
CAP-05	Intra-bank fund transfer	VERIFIED	E-18 `TransactionService.java` (35-46); E-33 `FundTransferService.java` (31-51)
CAP-06	Utility bill payment (ledger debit only)	VERIFIED	E-21 `TransactionService.java` (48-75); E-34 `UtilityPaymentService.java` (31-53)
CAP-07	Transfer / payment history (paged)	VERIFIED	E-33 `FundTransferService.java` (31-51); E-34 `UtilityPaymentService.java` (31-53)
CAP-08	Customer & user listing	VERIFIED	E-26 `core-banking-service/src/main/java/com/javatodev/finance/controller/*.java`; E-29 `UserService.java` (33-111)
CAP-09	Audit columns on channel records	VERIFIED	E-36 `*/src/main/java/com/javatodev/finance/model/dto/AuditAware.java`
CAP-10	Distributed tracing	VERIFIED	E-37 `*/build.gradle`; E-14 `[external] *-service.yml` (ddl-auto / management)
CAP-11	Notifications via RabbitMQ	DOCUMENTED (not implemented)	E-02 `README.md` (60-75); E-03 `README.md` (83-101)
CAP-12	Prometheus metrics / Kubernetes deployment	DOCUMENTED (not implemented)	E-03 `README.md` (83-101)

4. Core business workflows and user journeys

WF-01 Onboarding

1. Customer POST /user/api/v1/bank-users/register {email, identification, password}
2. User svc: Keycloak search(email) -> reject if found
3. User svc -> Core GET /api/v1/user/{identification}
4. Email must equal core email
5. Create disabled Keycloak user with password
6. Persist local user status=PENDING
7. Admin PATCH /update/{id} status=APPROVED -> Keycloak user enabled

Evidence: E-28 `UserController.java` (28-55); E-29 `UserService.java` (33-111)

WF-02 Login

1. Client calls Keycloak token endpoint (password grant)

2. Gateway validates JWT via JWKS on each call

Evidence: E-10 `realm-export.json` (686-873); E-15 `SecurityConfiguration.java` (24-35)

WF-03 Fund transfer

1. POST /fund-transfer/api/v1/transfer {fromAccount,toAccount,amount}
2. Persist fund_transfer PENDING
3. Feign -> Core POST /api/v1/transaction/fund-transfer
4. Core validates balance, updates both accounts, writes 2 ledger rows
5. Mark SUCCESS with core transactionId

Evidence: E-18 `TransactionService.java` (35-46); E-20 `TransactionService.java` (63-64,90-91,99-100); E-33 `FundTransferService.java` (31-51)

WF-04 Utility payment

1. POST /utility-payment/api/v1/utility-payment {providerId,amount,referenceNumber,account}
2. Persist PROCESSING
3. Feign -> Core util-payment: debit account, write ledger row
4. Mark SUCCESS

Evidence: E-21 `TransactionService.java` (48-75); E-34 `UtilityPaymentService.java` (31-53)

WF-05 History

1. GET /fund-transfer/api/v1/transfer or /utility-payment/api/v1/utility-payment (paged, all records)

Evidence: E-33 `FundTransferService.java` (31-51); E-34 `UtilityPaymentService.java` (31-53)

5. Business rules and validations

ID	Rule	Class	Evidence
BR-01	Registration requires the identification number to exist in core and email to match core record	VERIFIED	E-29 `UserService.java` (33-111)
BR-02	Email must not already exist in Keycloak (fuzzy search)	VERIFIED	E-29 `UserService.java` (33-111); E-30 `KeycloakUserService.java` (36)
BR-03	New online users start PENDING and disabled until APPROVED	VERIFIED	E-29 `UserService.java` (33-111)
BR-04	Debit allowed if actualBalance >= amount and actualBalance >= 0	VERIFIED	E-19 `TransactionService.java` (77-81)
BR-05	Amount must be positive	MISSING (not enforced)	E-19 `TransactionService.java` (77-81)
BR-06	Only ACTIVE accounts may transact	MISSING (enum exists, not enforced)	E-19 `TransactionService.java` (77-81); E-25 `core-banking-service/src/main/java/com/javatodev/finance/model/{AccountType,AccountStatus,TransactionType}.java`
BR-07	Customer may only debit own accounts	MISSING	E-33 `FundTransferService.java` (31-51)
BR-08	Transaction limits / cut-offs / fees	MISSING	No repository evidence available (absence of artefact)

6. Data entities and relationships

Entity / table	Owner service	Key attributes	Relationships
banking_core_user	core	id, first_name, last_name, email, identification_number	1-* account

Entity / table	Owner service	Key attributes	Relationships
banking_core_account	core	id, number, type, status, actual_balance, available_balance DECIMAL(19,2), user_id	*-1 user; 1-* transaction
banking_core_transaction	core	id, amount, transaction_type, reference_number, transaction_id, account_id	*-1 account (mapped @OneToOne)
banking_core_utility_account	core	id, number, provider_name	-
user	user-service	id, auth_id, identification, status + audit	auth_id -> Keycloak user
fund_transfer	fund-transfer	id, transaction_reference, from_account, to_account, amount, status + audit	transaction_reference -> core transaction_id
utility_payment	utility-payment	id, provider_id, amount, reference_number, account, transaction_id, status + audit	transaction_id -> core

7. API capabilities

Service	Method	Path	Purpose
Gateway	*	/user/ **/fund-transfer/** /banking-core/ **/utility-payment/**	Routing; JWT required except register & actuator
User	POST	/api/v1/bank-users/register	Register (public)
User	PATCH	/api/v1/bank-users/update/{id}	Change status
User	GET	/api/v1/bank-users, /api/v1/bank-users/{id}	List / read
Fund Transfer	POST/GET	/api/v1/transfer	Transfer / history
Utility Payment	POST/GET	/api/v1/utility-payment	Pay / history
Core	GET	/api/v1/account/bank-account/{n}, /api/v1/account/util-account/{provider}	Lookups
Core	POST	/api/v1/transaction/fund-transfer, /api/v1/transaction/util-payment	Ledger postings
Core	GET	/api/v1/user/{identification}, /api/v1/user	Customer lookup

8. Product modules and components

Module	Port	Java files	LOC	Responsibility
core-banking-service	8092	36	831	Simulated core ledger
internet-banking-user-service	8083	32	822	Registration, approval, Keycloak sync
internet-banking-fund-transfer-service	8084	25	547	Fund transfer orchestration + history
internet-banking-utility-payment-service	8085	23	530	Utility payment orchestration + history
internet-banking-api-gateway	8082	3	86	Edge routing + JWT validation
internet-banking-config-server	8090	1	15	Spring Cloud Config (git)
internet-banking-service-registry	8081	1	15	Eureka server

9. External integrations and product dependencies

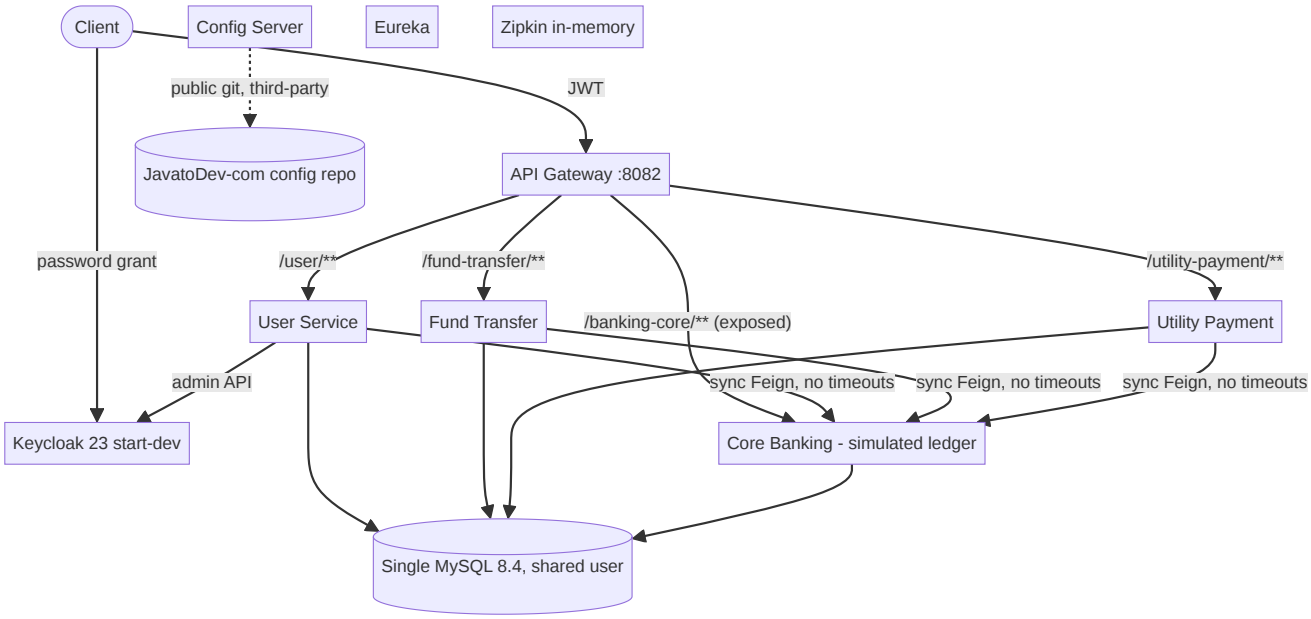
Integration	Used by	Class	Evidence
Keycloak Admin REST API	User service (client_credentials)	VERIFIED	E-31 `KeycloakProperties.java` (25-40)
Keycloak OIDC / JWKS	Gateway	VERIFIED	E-15 `SecurityConfiguration.java` (24-35)
External config repo (GitHub)	Config Server	VERIFIED	E-11 `application.yml` (1-10)
Zipkin	All services	VERIFIED	E-14 `[external] *-service.yml` (ddl-auto / management)

Integration	Used by	Class	Evidence
RabbitMQ	-	DOCUMENTED only	E-03 `README.md` (83-101)
Real CBS / payment rails / biller aggregator	-	MISSING	E-21 `TransactionService.java` (48-75)

10. Deployment and operational context

docker-compose on a single host: 7 app containers + MySQL + Keycloak + Postgres + Zipkin, static IPs on 172.25.0.0/16, images referenced as javatodev/*; services wait for registry/config/MySQL via wait-for-it.sh. No Kubernetes/IaC/CI.

Class: VERIFIED. Evidence: E-05 `docker-compose.yml` (1-139); E-39 `*/Dockerfile`; E-43 `git log`



11. Missing information and items requiring stakeholder validation

- Business KPIs, customers, revenue, adoption, SLAs - not in repository (STAKEHOLDER)
- Production deployment history / incidents - not in repository (STAKEHOLDER)
- Ownership of external config repo and Docker Hub images (STAKEHOLDER)
- Target core banking system and payment rails (STAKEHOLDER)
- Non-functional requirements: TPS, latency, RPO/RTO, data residency (MISSING)
- Security testing / VAPT reports (MISSING)
- Build/test execution results - dependency download blocked (HTTP 429) in assessment environment (LIMITATION)

Repository Inventory

Discovery checklist

Area	Status	Notes
Repository structure	VERIFIED	7 Gradle projects + docker-compose/, postman_collection/, draw_io/
README / documentation	VERIFIED	README.md only; links to external tutorial articles
Application source code	VERIFIED	121 Java files, ~2,846 LOC
Configuration files	VERIFIED	bootstrap*.yml in repo; runtime config in external repo (E-11, E-13, E-14)
Build & deployment files	VERIFIED	build.gradle x7, Dockerfile x7, docker-compose x2
Dependency management	VERIFIED	Gradle + Spring dependency-management BOM; no lock files, no version catalog
API specifications	VERIFIED (runtime only)	springdoc annotations; no static OpenAPI file; Postman collection

Area	Status	Notes
Database schemas & migrations	VERIFIED	Flyway in core only; Hibernate ddl-auto update elsewhere
Infrastructure-as-Code	MISSING	No Terraform/Helm/K8s manifests (README lists Kubernetes)
CI/CD pipelines	MISSING	Workflows removed in HEAD commit
Unit / integration / E2E tests	PARTIAL	18 unit tests (core); smoke tests; no integration/E2E automation
Architecture diagrams	PARTIAL	One draw.io diagram (centralised config); README links external image
Operational / monitoring config	PARTIAL	Actuator + Zipkin tracing; no metrics backend/alerts/dashboards
Security / authentication	VERIFIED	Keycloak realm export, gateway resource server, user-service admin client

Languages

- Java 21 (VERIFIED, build.gradle sourceCompatibility)
- SQL (Flyway migrations)
- YAML (Spring config, docker-compose)
- Shell (wait-for-it.sh)

Frameworks

- Spring Boot 3.2.4
- Spring Cloud 2023.0.0 (Gateway, Netflix Eureka, Config, OpenFeign)
- Spring Data JPA / Hibernate
- Spring Security OAuth2 Resource Server (gateway only)
- Micrometer Tracing (Brave) + Zipkin reporter
- springdoc-openapi 2.1.0
- Lombok
- Flyway 10.12.0 (core only)

Platforms

- MySQL 8.4.0
- PostgreSQL 15 (Keycloak)
- Keycloak 23.0.7 (+ admin client 24.0.4)
- Zipkin 3
- Docker / docker-compose 3.6
- Eclipse Temurin 21 JRE Alpine

Tools

- Gradle 8.6 wrapper x7
- Postman collection + environment
- draw.io diagram
- IntelliJ HTTP file (api_test.http)

Modules

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core-banking-service	8092	36	831	Simulated core ledger
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internet-banking-api-gateway	8082	3	86	Edge routing + JWT validation
internet-banking-config-server	8090	1	15	Spring Cloud Config (git)
internet-banking-service-registry	8081	1	15	Eureka server

Totals: ~2,846 production Java LOC in 121 files; ~413 test LOC in 10 files; 194 commits (2021-05-09 to 2026).

Evidence Index

All findings cite these IDs. Links point to the baseline commit `fa1c445`. Items prefixed `[external]` are in the externally-hosted config repository (github.com/JavatoDev-com/internet-banking-microservices-configurations (configuration/, branch main)).

ID	Path / source	Lines	What it shows
E-01	README.md	1-20	Self-description as tutorial 'Internet banking API concept'; releases 1.0.0 / 2.0.0
E-02	README.md	60-75	Test credentials published; list of microservices incl. RabbitMQ + Notification service 'PENDING Development'
E-03	README.md	83-101	Technology stack list (incl. RabbitMQ, Prometheus, Kubernetes, Sleuth)
E-04	LICENSE	1-3	MIT licence, (c) 2022 javatodev.com
E-05	docker-compose/docker-compose.yml	1-139	Full stack: Zipkin, Keycloak 23.0.7 start-dev, Postgres 15, MySQL, 7 services, static IPs 172.25.0.0/16, host ports published
E-06	docker-compose/docker-compose.yml	21-23,43,52	Hard-coded Keycloak admin/DB and MySQL root passwords
E-07	docker-compose/mysql/privileges.sql	1-2	Shared app DB user with CREATE/ALTER/DROP/DML on .; password in clear
E-08	docker-compose/mysql/privileges.sql	5-8	Four logical schemas on one MySQL instance
E-09	docker-compose/keycloak/realm-export.json	29,38	sslRequired=external; bruteForceProtected=false
E-10	docker-compose/keycloak/realm-export.json	686-873	Business clients with directAccessGrantsEnabled=true, wildcard redirect URIs '/*', committed client secrets (lines 759, 861)
E-11	internet-banking-config-server/src/main/resources/application.yml	1-10	Config Server git backend = third-party public repo JavatoDev-com/internet-banking-microservices-configurations
E-12	*/src/main/resources/bootstrap*.yml	-	Only app name + config URI per profile; default profile hard-codes 192.168.1.5
E-13	[external] internet-banking-api-gateway*.yml	routes	Gateway routes /user/**, /fund-transfer/**, /banking-core/**, /utility-payment/** (StripPrefix=1, lb://)
E-14	[external] *-service.yml	ddl-auto / management	ddl-auto: update for user, fund-transfer, utility-payment; actuator exposure '*' for gateway & fund-transfer
E-15	internet-banking-api-gateway/src/main/java/com/javatodev/finance/configuration/security/SecurityConfiguration.java	24-35	permitAll on register + all actuator paths; anyExchange().authenticated(); CSRF disabled; JWT via JWKS
E-16	internet-banking-api-gateway/src/main/java/com/javatodev/finance/configuration/GatewayConfiguration.java	18-28	GlobalFilter builds mutated request with X-Auth-Id but returns original exchange
E-17	internet-banking-user-service/src/main/java/com/javatodev/finance/configuration/filter/AppAuthUserFilter.java	21-33	Services trust X-Auth-Id header for audit identity; no JWT validation downstream
E-18	core-banking-service/src/main/java/com/javatodev/finance/service/TransactionService.java	35-46	fundTransfer: reads accounts, validateBalance, internalFundTransfer
E-19	core-banking-service/src/main/java/com/javatodev/finance/service/TransactionService.java	77-81	validateBalance only checks balance<0 or balance0, no status check)
E-20	core-banking-service/src/main/java/com/javatodev/finance/service/TransactionService.java	63-64,90-91,99-100	availableBalance recomputed from already-updated actualBalance -> double adjustment

ID	Path / source	Lines	What it shows
E-21	core-banking-service/src/main/java/com/javatodev/finance/service/TransactionService.java	48-75	utilPayment: debit only; comment 'we can call third party API...'; no settlement credit
E-22	core-banking-service/src/main/java/com/javatodev/finance/model/entity/TransactionEntity.java	17-35	Ledger row: amount, type, reference, transactionId; @OneToOne(cascade=ALL) to account; no timestamp
E-23	core-banking-service/src/main/java/com/javatodev/finance/model/entity/BankAccountEntity.java	16-37	Mutable balances; no @Version
E-24	core-banking-service/src/main/resources/db/migration/	V1.0.*	Flyway schema (user, account, utility_account, transaction) + temp_data seed migration
E-25	core-banking-service/src/main/java/com/javatodev/finance/model/{AccountType,AccountStatus,TransactionType}.java	-	Domain enums
E-26	core-banking-service/src/main/java/com/javatodev/finance/controller/*.java	-	Core APIs: account lookup, fund-transfer, util-payment, user lookup
E-27	*/src/main/java/com/javatodev/finance/exception/GlobalExceptionHandler.java	core:27 ft:24 user:27 up:27	Generic handler returns 400 'Exception occur inside API ' + e
E-28	internet-banking-user-service/src/main/java/com/javatodev/finance/controller/UserController.java	28-55	register (logs request.toString incl. password, line 31), PATCH update status (no role check), list, read
E-29	internet-banking-user-service/src/main/java/com/javatodev/finance/service/UserService.java	33-111	Registration flow (Keycloak search, core lookup, create disabled user, get(0) line 67); readUsers N+1 (line 84); approve enables Keycloak user
E-30	internet-banking-user-service/src/main/java/com/javatodev/finance/service/KeycloakUserService.java	36	users().search(email) - fuzzy search
E-31	internet-banking-user-service/src/main/java/com/javatodev/finance/configuration/keycloak/KeycloakProperties.java	25-40	Static lazily-initialised Keycloak client (client_credentials)
E-32	internet-banking-user-service/src/main/java/com/javatodev/finance/model/dto/User.java	1-20	@Data DTO including password field
E-33	internet-banking-fund-transfer-service/src/main/java/com/javatodev/finance/service/FundTransferService.java	31-51	Save PENDING -> Feign core -> SUCCESS; no failure handling; findAll unscoped (line 50)
E-34	internet-banking-utility-payment-service/src/main/java/com/javatodev/finance/service/UtilityPaymentService.java	31-53	Save PROCESSING -> Feign core -> SUCCESS; no failure handling; findAll unscoped (line 50)
E-35	internet-banking-fund-transfer-service/src/main/java/com/javatodev/finance/configuration/CustomFeignClientConfiguration.java	13	Feign Logger.Level.FULL
E-36	*/src/main/java/com/javatodev/finance/model/dto/AuditAware.java	-	createdBy/Date, modifiedBy/Date, @Version on channel entities
E-37	*/build.gradle	-	Spring Boot 3.2.4, Spring Cloud 2023.0.0, springdoc webflux-ui 2.1.0, Keycloak admin 24.0.4, Flyway 10.12.0, MySQL connector 8.4.0
E-38	*/gradle/wrapper/gradle-wrapper.properties	-	7 independent Gradle 8.6 builds; no root project
E-39	*/Dockerfile	-	eclipse-temurin:21.0.2_13-jre-alpine; ADD prebuilt jar; no USER; wait-for-it.sh
E-40	*/src/test/**	-	18 unit tests in core-banking (Transaction 9, Account 6, User 3); contextLoads only elsewhere
E-41	internet-banking-user-service/src/test/resources/application.yml	35	Keycloak client secret committed in test config

ID	Path / source	Lines	What it shows
E-42	postman_collection/JAVA_TO_DEV_MICROSERVICES.postman_collection.json	-	14 requests; uses /core/ **and /payment/** and /bank-user paths that do not match gateway routes / controller paths
E-43	git log	-	194 commits 2021-05-09..2026; last commit removed GitHub Actions workflows; no .github/workflows present
E-44	*/src/main/java/com/javatodev/finance/model/*/AccountResponse.java	9	Account number typed Long in fund-transfer & utility-payment vs String in core
E-45	core-banking-service/src/main/resources/messages*.properties	-	i18n: default + French messages only
E-46	draw_io/centralized_configuration.drawio	-	Only architecture diagram in repo (centralised configuration)
E-47	*/src/main/java/com/javatodev/finance/controller/*Controller.java	-	springdoc @Tag/@Operation annotations (OpenAPI generated at runtime; no static spec file)
E-48	.github/FUNDING.yml	-	Patreon funding file of original author

Limitations and information gaps

- Static analysis only - services were not executed end-to-end; findings marked Medium confidence (e.g. TDD-ARC-07) should be confirmed at runtime.
- `./gradlew test` with JDK 21 could not complete: Maven Central returned HTTP 429 (rate limit) during dependency resolution in the assessment environment. Test pass/fail status and dependency CVE scan are therefore not available.
- Runtime configuration lives in an external, third-party repository; it was reviewed at its current HEAD, which the services do not pin.
- Technology lifecycle/support statements were not re-verified online (confidence Medium).
- Business data (customers, revenue, KPIs, market share, adoption, SLAs, incidents) is not in the repository and has not been estimated.
- The 35-item capability reference list and effort/timeline estimates are assessor judgement, not market or benchmark data.
- No UI exists; accessibility could not be assessed.